

SUMMARY

SPARK addresses a fundamental problem in child development: **millions of children at risk of learning difficulties are identified only after those challenges have become entrenched**, increasing the likelihood of persistent academic underachievement, mental health challenges, and downstream **pressure on education, health, social service, and justice systems**. Grounded in the hypothesis that learning difficulties are preceded by measurable neurobehavioral warning signs in early childhood, SPARK will combine longitudinal data analysis, machine learning methods, and rigorous validation to **build a comprehensive atlas** linking early neurodevelopmental signals to behavioral indicators. This atlas will provide the foundation **for scalable early-risk screeners** that **enable earlier identification, intervention**, and improved long-term developmental and educational outcomes.

THE CHALLENGE

The problem

An estimated **300 million children^[1]** worldwide are **affected by learning difficulties**, with a long-term economic **cost of US\$10 trillion^[2]** in lost lifetime productivity in the US alone, about **9% of current world GDP**. The cost of late identification extends far beyond schools. Learning-related vulnerabilities are overrepresented in the systems that carry the burden of missed support: **one in four adults in U.S. prisons** report a cognitive disability^[3], and about **one in four adults experiencing homelessness screen positive** for cognitive impairment^[4]. Early learning risk is a societal challenge that places long-term pressure on **education, health, social service, homelessness, and justice systems**.

A central failure is the **system's inability to detect vulnerability early and reliably**. Most education systems operate on a **wait-to-fail model**: children are diagnosed only after difficulties become visible through underachievement or behavioral disruption. Even then, identification is highly variable and often imprecise, relying on resource-intensive, specialist-led processes that are difficult to scale and unevenly available. As a result, many children are identified too late, and **most are never identified at all**. By the time these challenges are diagnosed, the period in which intervention is most effective has already narrowed, and the consequences of missed support have begun to accumulate academically, emotionally, and behaviorally.

What is missing is a standardized, accessible & practical bridge between early neurodevelopmental signals and scalable behavioral indicators. SPARK will build that bridge by translating robust, cross-validated evidence into actionable **screening tools that can support timely educational and clinical response**.

Why now

SPARK is possible now because **three major capabilities** have recently matured, making this breakthrough possible for the first time. First, **advances in neuroscience** have made it increasingly feasible to reliably detect and interpret meaningful neurodevelopmental signals in early childhood. Second, **digital behavioral tools** can now generate large-scale, high-resolution behavioral datasets on how every child learns. Third, **advances in artificial intelligence** can link neural signatures to behavioral patterns, enabling predictive models that can identify underlying learning risks without requiring EEG at scale. Many datasets from different global studies are now "open," but have not been integrated, or cross-validated, to enable practical tool development.

Currently, **no existing organization has the mandate or expertise to integrate neuroscience, large-scale behavioral data, and machine learning** into early, scalable screening tools that are accessible and adaptable to different cultural and social settings. Academia is fragmented and risk-averse; venture-backed startups cannot justify high-risk neuroscience research; large education markets reward engagement more than evidence generation; and clinical pathways are limited by cost and regulation. **SPARK brings together** the scientific, technical, and implementation **experience needed to** turn these converging capabilities into **validated tools that can be applied, adapted, and scaled across diverse global contexts**.

THE OPPORTUNITY

The Thesis

The developing brain can exhibit early neurodevelopmental signatures that indicate elevated risk for learning difficulties. By using longitudinal data and artificial intelligence, those early signals can be transformed into scalable behavioral screening tools that identify risk sooner, enable timely intervention, and improve continuously over time.

This thesis contains four claims:

Claim 1 — Early risk is measurable before conventional school failure emerges^{[5][6]}.

Claim 2 — That vulnerability can be captured through integrated measurement of **neurophysiological signals** (such as EEG) and behavior, and **modeled with machine learning**^[7].

Claim 3 — These models can be **translated into behavioral proxies to triage and support scalable screening** without requiring EEG or specialist equipment in the final deployment setting^[8].

Claim 4 — Once deployed and linked to longitudinal follow-up, these tools can **improve iteratively** becoming part of a **preventive learning infrastructure** rather than a one-time assessment tool.

Focus Areas

SPARK will be organized around four integrated focus areas that together build off existing data and tools to define the pathway from early signal discovery to scalable screening:

1. **Biomarker discovery:** Identify candidate EEG signatures associated with vulnerability to later learning risk during the critical developmental window of ages 2 to 6.
2. **Behavioral proxy development:** design and refine child-friendly tasks that capture observable patterns associated with neurodevelopmental risk and generate scalable behavioral indicators.
3. **Longitudinal validation:** follow children over time to determine which early neurophysiological and behavioral patterns reliably predict later developmental and educational trajectories.
4. **Screening tool translation:** convert validated signals and behavioral proxies into practical, low-cost screening tools suitable for educational, clinical, and community settings.

Equity by design: Across all workstreams, SPARK prioritizes robustness across language, culture, socioeconomic context, and developmental opportunity.

KEY STAKEHOLDERS

SPARK will leverage and expand a coordinated stakeholder ecosystem built from over a decade of applied research and delivery, that evolves from discovery to implementation and scale. **Phase 1: Research and measurement partners.** Partnerships with the University of Minnesota's Institute of Child Development, Tecnológico de Monterrey, and the Universidad de Montevideo provide the scientific foundation for biomarker discovery, EEG-based measurement, signal interpretation, and validation. **Phase 2: Community and implementation partners.** An existing network of 4,000+ health professionals across 12+ countries will enable multi-site testing, longitudinal follow-up, participant engagement, and large-scale cross-validation across schools, early childhood systems, and community settings. **Phase 3: Public-sector and multilateral pathways to scale.** Current government partnerships in Mexico, Uruguay, Chile and multiple U.S. states create pathways for adoption, sustainability and policy alignment, with a current reach of +6 million children. In addition to **core philanthropy funding** for SPARK's development, a longstanding, successful relationship with the Inter-American Development Bank will support regional dissemination, institutional coordination, and cross-country scaling.

BUDGET

SPARK is organized as a **philanthropy-funded research organization** because this work **requires integrated execution** across discovery, validation, and translation. The program is proposed as a **five-year, time-bound effort with measurable milestones** and a **US\$20 million budget** to build the internal scientific, technical, and operational capacity required, while leveraging external partners for specialized measurement, validation, and scale. From the outset, SPARK is designed for **long-term sustainability** through **public-sector adoption across jurisdictions**, with future opportunities for continued technology development, deployment, and earned-revenue models beyond the philanthropic phase.

Budget	%	Amount
Core team & operations	35%	\$7M
Scientific and measurement infrastructure	20%	\$4M
Screening ecosystem and software development	20%	\$4M
Field deployment & longitudinal validation	15%	\$3M
External partnerships, adaptation, and deployment readiness	10%	\$2M

TEAM AND ROLES

Agustín Pardo Van Thienen — Founder and CEO

I bring more than **15 years of extensive experience** designing and deploying adaptive learning systems for young children. I have repeatedly encountered the challenge that now defines SPARK: children at greatest risk are identified too late, after the period when intervention could have the greatest impact.

I have built **adaptive learning products, behavioral measurement systems, and AI-enabled data pipelines**, and I have taken them from concept to large-scale implementation through partnerships with governments, researchers, and multilaterals. Through my company, Wumbox, I led a high-risk early literacy initiative supported by the **Inter-American Development Bank** that reached more than **250,000 students**, expanded it beyond its original scope, and produced a rare early repayment success for the Bank. Since then, I have helped lead programs, research partnerships, and government implementation across **more than 20 countries, reaching more than 10 million beneficiaries**. I bring to SPARK not only product and technical experience, but also the ability to align researchers, development teams, governments, and field networks around launch, validation and execution - all core components needed to translate early neurodevelopmental signals into scalable screening tools.

Dr. Stephanie Carlson (University of Minnesota's Institute of Child Development) is a globally respected leader on executive function development in early childhood, (central to our biomarker discovery workstream) and will act as our **founding Scientific Advisor**.



Dr. Bruce McCandliss (Stanford University) is a renowned leader in educational cognitive neuroscience, linking brain-based learning science to classroom development and will act as our **Scientific Advisor** (in conversation)



Additional roles in Founding Team: Director of Behavioral Assessment · Director of Data Science · Director of Strategy, Implementation & Alliances · Director of Intervention Translation · COO/CFO

Advisory and collaboration network: Our Advisory board will bring together experts in brain development, child development, psychometrics, education systems, ethics, and user-focused public-sector adoption. SPARK's development has already been informed by expert advisors and collaborators across these domains and the broader translational arena. This includes academic and cross-sector leaders such as Dr. Jane Roskams (Ren Phil), Dr. Stanislas Dehaene (CdF), Dr. Pedro Valdes-Sosa (UESTC), Dr. Adele Diamond (UBC), Dr. Adam Gazzaley (UCSF), Dr. Maki Koyama (CUNY), Dr. Sharon Vaughn (UT Austin), Dr. Nancy Jordan (UD), Dr. Brian Butterworth (UCL), Dr. Daniel Ansari (UBC), Dr. Philip Zelazo (ICD).

IMPACT AND OUTCOMES

SPARK's **North Star** is the large-scale deployment of validated early-risk screening infrastructure through educational, clinical, and public-sector systems, building that future in iterative steps:

Horizon	Timeline	Deliverables
Phase 1: Build	Year 1-2	Build SPARK, launch multimodal data pipeline, deploy first-generation screeners, identify and validate initial signal-to-proxy candidates.
Phase 2: Validate	Year 3	Refined biomarker candidates; second-generation behavioral proxies; longitudinal validation cohort established; pre-registered longitudinal/RCT evidence study launched.
Phase 3: Scale	Year 4-5	Validated Atlas ; aligned institutional partnerships; third-generation screeners deployed, with up to 10M children screened and risk identified 5–7 years earlier .

From waiting for children to fail, to understanding them early enough to help. That's what SPARK is for.

APPENDIX

¹ Bozatlı L, et al. *Children at Risk of Specific Learning Disorder: A Study on Prevalence and Risk Factors*. Children (Basel). 2024;11(7):759. [PubMed](#) | [Article](#)

² Hoover Institution. The Pandemic in Perspective. 2025; NCES. Students With Disabilities. 2024; APA. Specific Learning Disorder. SPARK's **US\$10 trillion** figure is a conservative modeled estimate: US\$90T total U.S. learning-loss exposure × ~10–12% plausible learning-difference-related share. This share is supported by official SLD identification rates and broader 5–15% prevalence estimates for learning disabilities. [Stanford Hoover Institution](#) | [Press release](#)

World Bank. *Foundational Learning*. 2025. SPARK's **US\$4 trillion** figure is a conservative modeled estimate derived from the World Bank's **US\$21 trillion** lifetime-earnings estimate for the broader learning crisis. [World Bank](#) | [Press release](#)

³ Maruschak, Bronson, & Alper (2021), *Disabilities Reported by Prisoners*. It reports that among all state and federal prisoners in 2016, 22.8% reported a cognitive disability and **24.4% reported having been told they had an attention deficit disorder**. This is the DOJ/BJA publication you were referring to. [Department of Justice](#) | [Publication](#)

⁴ Depp, Vella, Orff, & Twamley (2015), *A Quantitative Review of Cognitive Functioning in Homeless Adults*. That review states that in a pooled sample of 2,969 homeless adults, **about 25% screened positive for cognitive impairment**. [NIH](#) | [Article](#)

⁵ Gaab N, et al. *The Reading Brain from Womb to Classroom: Typical and Atypical Development and Implications for a Preventative Educational Model*. 2026. [PsyArXiv](#) | [Article](#)

⁶ Eng CM, et al. *Infant attention and frontal EEG neuromarkers of childhood ADHD*. 2025. [PubMed](#) | [Article](#)

⁷ Nakai T, et al. *From brain to education through machine learning: Predicting literacy and numeracy skills from neuroimaging data*. 2024. [PubMed](#) | [Article](#)

⁸ Liu YS, et al. *Early identification of children with Attention-Deficit/Hyperactivity Disorder (ADHD)*. 2024. [PubMed](#) | [Article](#)